



Jérôme Kasparian
Professeur

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To whom it may concern

Alexis Gomel has been working as an assistant and PhD candidate in my group, in the Department of applied physics of the University of Geneva, from May 2018 to November 2022. His research work focused on relations between non-linear systems, especially non-linear optics (laser filamentation) and hydrodynamics (optical solitons in particular), and the characterization of tipping points. In this framework he performed experimental work both in the laboratory in Geneva and at a large wave tank facility in Sydney, theoretical work, as well as numerical simulations. His ability to tackle different subjects and master a wide range of technical approaches in his research is impressing.

Although the Covid situation limited the amount of experiments he was able to perform, especially due to travel restrictions to the wave tank facility in Sydney, he obtained a very high number of results, which resulted in four published articles, from which three are as first author, including one in Physical Review Letters.

Alexis is very rigorous and hard working. During his PhD, he developed an excellent creativity together with the ability to always seek for deep understanding of the systems he is working on. Furthermore, he integrated very fast in my group as well as with external collaborators, proving his ability to work, share ideas and collaborate in a highly multicultural environment. For example, he contributed to the initiative to revive a PhD students' seminar in the Department of Applied Physics when the Covid restrictions ended. He also gained a high level of autonomy in his work, allowing him to make initiatives and pursue his research by himself when relevant.

Finally, Alexis was involved in teaching laboratory courses for first year students, and in the master course on Nonlinear Systems, which I am giving. In both cases, his rigor, commitment, and enthusiasm was greatly appreciated by both students and the instructors.

Finally, I would like to underline the high teamwork abilities and human qualities of Alexis, which always makes it a pleasure to work with him.

For all these reasons, I am happy to strongly support Alexis and feel he would be a highly valuable asset for any organization.

I remain at your disposal for any further information you may need.

Best regards,

Prof. Jérôme Kasparian